Note 1

Click Or enter the below link in the URL to download datasets https://drive.google.com/file/d/1ezMRnPMhoJZbtIMHXDPyZLYGsnbQYMNx/view?usp=sharing

1. Consider Iris dataset and implement Decission tree and perform pruning as needed.

   You can load the iris dataset using the code below -

   ```
   from sklearn import datasets
   iris = datasets.load_iris()
   ```

2. Implement the random forest algorithm with n decision trees.

   (a) Use any of the below dataset -

      i. Sonar dataset

      ii. Ionosphere dataset

   (b) Divide the dataset for training and testing in 7:3 ratio.

   (c) Take the value of n such that $n > 15$.

   (d) Use the Following metrics for showing the performance of the model -

      i. Accuracy

      ii. Confusion Matrix

      iii. Preciosion

      iv. Recall

      v. F1-Score

      vi. Area Under the Curve (AUC)

   (e) Also, compare the results of the implemented random forest algorithm with the results of the available library function for the same.

(cleaning of dataset might be required before processing, like filling in the missing values and converting words to numbers)